

JKBOSE 2026

CLASS 11TH

TRIGONOMETRIC FUNCTIONS

MATHS

LECTURE-07

BY-MEHRAJ SIR



TOPICS TO BE COVERED

1

Exercise Discussion 3.3

$$\begin{aligned}
 * \cos x + \cos y &= 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \checkmark \\
 * \cos x - \cos y &= -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) \checkmark \\
 * \sin x + \sin y &= 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \checkmark \\
 * \sin x - \sin y &= 2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) \checkmark
 \end{aligned}$$

EXERCISE – 3.3

QUESTION



Prove the following:

$$\frac{\cos 9x - \cos 5x}{\sin 17x - \sin 3x} = -\frac{\sin 2x}{\cos 10x}$$

Proof: $\text{LHS} = \frac{\cos 9x - \cos 5x}{\sin 17x - \sin 3x}$

$$\Rightarrow \text{LHS} = \frac{-2 \sin\left(\frac{9x+5x}{2}\right) \sin\left(\frac{9x-5x}{2}\right)}{2 \cos\left(\frac{17x+3x}{2}\right) \sin\left(\frac{17x-3x}{2}\right)}$$

$$\Rightarrow \text{LHS} = -\frac{\cancel{\sin 7x} \sin 2x}{\cos 10x \cancel{\sin 7x}}$$

$$\Rightarrow \text{LHS} = -\frac{\sin 2x}{\cos 10x} = \text{RHS}$$

Hence proved.

QUESTION

Prove the following:

$$\frac{\sin 5x + \sin 3x}{\cos 5x + \cos 3x} = \tan 4x$$

Proof: LHS = $\frac{\sin 5x + \sin 3x}{\cos 5x + \cos 3x}$

$$\Rightarrow \text{LHS} = \frac{\cancel{2} \sin\left(\frac{5x+3x}{2}\right) \cos\left(\frac{5x-3x}{2}\right)}{\cancel{2} \cos\left(\frac{5x+3x}{2}\right) \cos\left(\frac{5x-3x}{2}\right)}$$

$$\Rightarrow \text{LHS} = \frac{\sin 4x \cancel{\cos x}}{\cos 4x \cancel{\cos x}}$$

$$\Rightarrow \text{LHS} = \tan 4x = \text{RHS}$$

Hence proved.

QUESTION

Prove the following:

$$\frac{\sin x - \sin y}{\cos x + \cos y} = \tan \frac{x - y}{2}$$

Proof: LHS = $\frac{\sin x - \sin y}{\cos x + \cos y}$

$$\Rightarrow \text{LHS} = \frac{2 \cancel{\cos\left(\frac{x+y}{2}\right)} \sin\left(\frac{x-y}{2}\right)}{2 \cancel{\cos\left(\frac{x+y}{2}\right)} \cos\left(\frac{x-y}{2}\right)}$$

$$\Rightarrow \text{LHS} = \tan\left(\frac{x-y}{2}\right) = \text{RHS}$$

hence proved.

QUESTION



Prove the following:

$$\frac{\sin x + \sin 3x}{\cos x + \cos 3x} = \tan 2x$$

Proof: LHS = $\frac{\sin x + \sin 3x}{\cos x + \cos 3x}$

$$\Rightarrow \text{LHS} = \frac{2 \sin\left(\frac{x+3x}{2}\right) \cos\left(\frac{x-3x}{2}\right)}{2 \cos\left(\frac{x+3x}{2}\right) \cos\left(\frac{x-3x}{2}\right)}$$

$$\Rightarrow \text{LHS} = \frac{\sin 2x}{\cos 2x} = \tan 2x = \text{RHS}$$

Hence proved.

QUESTION

$$\begin{aligned}\cos 2x &= \cos^2 x - \sin^2 x \\ \Rightarrow -\cos 2x &= \sin^2 x - \cos^2 x.\end{aligned}$$

$$\sin\left(\frac{x-3x}{2}\right) = \sin\left(\frac{-2x}{2}\right)$$



Prove the following:

$$\frac{\sin x - \sin 3x}{\sin^2 x - \cos^2 x} = 2\sin x$$

Proof: LHS = $\frac{\sin x - \sin 3x}{\sin^2 x - \cos^2 x}$

$$\Rightarrow \text{LHS} = \frac{2\cos\left(\frac{x+3x}{2}\right)\sin\left(\frac{x-3x}{2}\right)}{-\cos 2x}$$

$$\Rightarrow \text{LHS} = \frac{2\cancel{\cos 2x}\sin(-x)}{-\cancel{\cos 2x}}$$

$$\Rightarrow \text{LHS} = \frac{2x/\cancel{\sin x}}{\cancel{-1}}$$

$$\Rightarrow \text{LHS} = 2\sin x = \text{RHS}$$

Hence proved.

QUESTION

(Imp)

Prove the following:

$$\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x} = \cot 3x$$

Proof: LHS = $\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x}$

$$\Rightarrow \text{LHS} = \frac{(\cos 4x + \cos 2x) + \cos 3x}{(\sin 4x + \sin 2x) + \sin 3x}$$

$$\Rightarrow \text{LHS} = \frac{2 \cos\left(\frac{4x+2x}{2}\right) \cos\left(\frac{4x-2x}{2}\right) + \cos 3x}{2 \sin\left(\frac{4x+2x}{2}\right) \cos\left(\frac{4x-2x}{2}\right) + \sin 3x}$$

$$\Rightarrow \text{LHS} = \frac{2 \cos 3x \cos x + \cos 3x}{2 \sin 3x \cos x + \sin 3x}$$

$$\Rightarrow \text{LHS} = \frac{\cos 3x (2 \cos x + 1)}{\sin 3x (2 \cos x + 1)} = \frac{\cos 3x}{\sin 3x} = \cot 3x$$

$$\Rightarrow \boxed{\text{LHS} = \text{RHS}} \quad \text{Hence proved.}$$

QUESTION

Imp

$$\cot(x+y) = \frac{\cot x \cot y - 1}{\cot y + \cot x}$$



Prove the following:

$$\cot x \cot 2x - \cot 2x \cot 3x - \cot 3x \cot x = 1$$

Proof: LHS = $\cot x \cot 2x - \cot 2x \cot 3x - \cot 3x \cot x$

$$\Rightarrow \text{LHS} = \cot x \cot 2x - \cot 3x (\cot 2x + \cot x)$$

$$\Rightarrow \text{LHS} = \cot x \cot 2x - \cot(x+2x) (\cot 2x + \cot x)$$

$$\Rightarrow \text{LHS} = \cot x \cot 2x - \left[\frac{\cot x \cot 2x - 1}{\cot 2x + \cot x} \right] (\cot 2x + \cot x)$$

$$\Rightarrow \text{LHS} = \cancel{\cot x \cot 2x} - \cancel{\cot x \cot 2x} + 1$$

$$\Rightarrow \text{LHS} = 1 = \text{RHS.}$$

Hence proved.

QUESTION

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$



Prove the following:

$$\tan 4x = \frac{4 \tan x (1 - \tan^2 x)}{1 - 6 \tan^2 x + \tan^4 x}$$

Proof: LHS = $\tan(4x) = \tan 2(2x)$

$$\Rightarrow \text{LHS} = \frac{2 \tan(2x)}{1 - \tan^2(2x)}$$

$$\Rightarrow \text{LHS} = \frac{2 \left(\frac{2 \tan x}{1 - \tan^2 x} \right)}{1 - \left(\frac{2 \tan x}{1 - \tan^2 x} \right)^2} = \frac{(4 \tan x)}{(1 - \tan^2 x)} \left(\frac{(1 - \tan^2 x)^2 - 4 \tan^2 x}{(1 - \tan^2 x)^2} \right)$$

$$\Rightarrow \text{LHS} = \frac{4 \tan x}{\cancel{(1 - \tan^2 x)}} \times \frac{(1 - \tan^2 x)^2}{(1 - \tan^2 x)^2 - 4 \tan^2 x}$$

$$\Rightarrow \text{LHS} = \frac{4 \tan x (1 - \tan^2 x)}{1 + \tan^4 x - 2 \tan^2 x - 4 \tan^2 x} = \frac{4 \tan x (1 - \tan^2 x)}{1 - 6 \tan^2 x + \tan^4 x}$$

$$\Rightarrow \boxed{\text{LHS} = \text{RHS}}$$

Hence proved.

QUESTION

$$\cos 2x = 2\cos^2 x - 1 \\ 1 - 2\sin^2 x$$

Prove the following:

$$\cos 4x = 1 - 8\sin^2 x \cos^2 x$$

Proof: LHS = $\cos 4x = \cos 2(2x)$

$$\Rightarrow \text{LHS} = 2\cos^2(2x) - 1$$

$$\Rightarrow \text{LHS} = 2(2\cos^2 x - 1)^2 - 1$$

$$\Rightarrow \text{LHS} = 2(4\cos^4 x + 1 - 4\cos^2 x) - 1$$

$$\cos^2 x \times \cos^2 x$$

$$\Rightarrow \text{LHS} = 8\cos^4 x + 2 - 8\cos^2 x - 1$$

$$\Rightarrow \text{LHS} = 8(\cos^2 x)^2 - 8\cos^2 x + 1$$

$$\Rightarrow \text{LHS} = 8(1 - \sin^2 x)^2 - 8\cos^2 x + 1$$

$$\Rightarrow \text{LHS} = 8(1 + \sin^4 x - 2\sin^2 x) - 8\cos^2 x + 1$$

$$\Rightarrow \text{LHS} = 8 + 8\sin^4 x - 16\sin^2 x - 8\cos^2 x + 1$$

$$\Rightarrow \text{LHS} = 8 + 8\sin^2 x \times \sin^2 x - 16\sin^2 x - 8\cos^2 x + 1$$

$$\Rightarrow \text{LHS} = 9 + 8\sin^2 x(1 - \cos^2 x) - 16\sin^2 x - 8\cos^2 x$$

$$\text{LHS} = 9 + 8\sin^2 x (1 - \cos^2 x) - 16\sin^2 x - 8\cos^2 x$$

$$\Rightarrow \text{LHS} = 9 + 8\sin^2 x - 8\sin^2 x \cos^2 x - 16\sin^2 x - 8\cos^2 x$$

$$\Rightarrow \text{LHS} = 9 - 8\sin^2 x - 8\cos^2 x - 8\sin^2 x \cos^2 x$$

$$\Rightarrow \text{LHS} = 9 - 8(\sin^2 x + \cos^2 x) - 8\sin^2 x \cos^2 x$$

$$\Rightarrow \text{LHS} = 9 - 8 - 8\sin^2 x \cos^2 x$$

$$\Rightarrow \text{LHS} = 1 - 8\sin^2 x \cos^2 x = \text{RHS}$$

Hence proved.

QUESTION

$$\cos 3x = 4\cos^3 x - 3\cos x$$

$$\cos 2x = 2\cos^2 x - 1$$

Prove the following:

$$\cos 6x = 32\cos^6 x - 48\cos^4 x + 18\cos^2 x - 1$$

Proof: LHS = $\cos 6x = \cos 3(2x)$

$$\Rightarrow \text{LHS} = 4\cos^3 2x - 3\cos 2x$$

$$\Rightarrow \text{LHS} = 4(2\cos^2 x - 1)^3 - 3(2\cos^2 x - 1)$$

$$\Rightarrow \text{LHS} = 4(8\cos^6 x - 1 - 3 \times 4\cos^4 x + 3 \times 2\cos^2 x) - 6\cos^2 x + 3$$

$$\Rightarrow \text{LHS} = 32\cos^6 x - 48\cos^4 x + 24\cos^2 x - 4 - 6\cos^2 x + 3$$

$$\Rightarrow \text{LHS} = 32\cos^6 x - 48\cos^4 x + 18\cos^2 x - 1$$

$$\Rightarrow \boxed{\text{LHS} = \text{RHS}}$$

Hence proved.

Thank You

* Trust ✓

* Panic X.

* lecture ko one time
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koshish.

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